

Analysis of the effect in the electricity market equilibrium of a wind-power plant

Energy Markets - 2on Laboratory Assignment

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Abstract—This report presents a study of a small energy distribution-generation-consumption network and examines the impact in market clearing and power flow of adding a wind power plant. GAMS (General Algebraic Modeling System) is used to find the market equilibrium for both scenarios.

Keywords: Energy Distribution; Energy Generation; Energy Consumption; Wind Power Plant Integration; Market Equilibrium; Day-Ahead Market; Renewable Energy; GAMS; Electricity Market Analysis; TCMA.

I. INTRODUCTION

A simplified representation of the day-ahead electricity market of a fictitious country is used to analyze the optimization problem, modeling it as a TCMA (Transmission Constrained Multi-Period Auction Model).

The proposed fictitious market has 4 buses, 3 Demand points, 5 Generation points and 5 power lines is organized as seen in Fig 1.

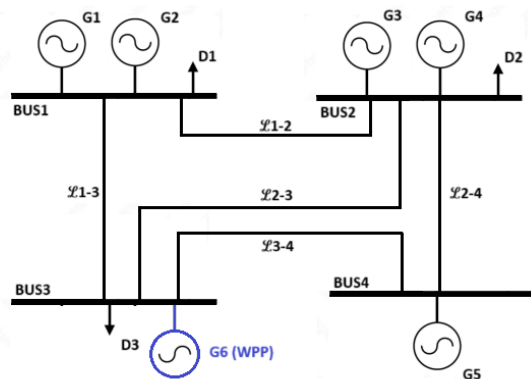


Fig 1. Single line diagram of the studied network (WPP in blue)

The special case in which an extra WPP is added in the bus3 is highlighted in blue.

A. Objectives of the study

The goal is to find the equilibrium points of the base case, which is the system without the wind power plant (WPP). After that, the WPP will be introduced at bus 3, and the equilibrium points identified by GAMS for both scenarios will be compared.

The key parameters for this comparison will be nodal prices, traded energy at each bus and time period, aggregated generation and demand functions, powerline congestions, economic transactions of producers and consumers, and merchandising surplus.

B. Structure of the report

The report begins with a description of both model's characteristics, with and without the WPP (Single-line diagram).

Next, point II explains in detail the data and how is translated into the TCMA mathematical model.

Following with point III, the results obtained by GAMS are presented. A MATLAB script is used to visualize the different market clearings for each bus and each time period.

Finally, the report ends with IV with brief conclusions and references.

II. DESCRIPTION OF THE PROBLEM AND MATHEMATICAL MODEL

A. Presenting model characteristics

All 5 Generation units are modeled following the data provided in Table1. Each Generator offers two different blocks (indexed as k) with its corresponding power in MW and price in €/Mwh. Limitations such as maximum/minimum powers and ramp up/down parameters are also provided; those define real market constraints.

Table 1, Base Model – Producers Characteristics

Generation units	$P_{G_i}^{max}$	$P_{G_i}^{min}$	$R_{G_i}^{up}$	$R_{G_i}^{down}$	$P_{G_i}^0$	$P_{G_{ikt}}^b [MW] \forall t$		$\lambda_{G_{ikt}}^b \left[\frac{\text{€}}{MWh} \right] \forall t$	
	[MW]	[MW]	[MW]	[MW]	[MW]	$k=1$	$k=2$	$k=1$	$k=2$
1	300	30	100	100	150	100	200	10	30
2	275	28	30	30	28	50	25	40	65
3	100	10	25	20	50	30	70	40	59
4	50	5	10	10	25	30	20	40	67
5	200	20	50	50	100	100	0	30	30

The extra WPP (wind power plant) characteristics is modeled following the data in Table2. This new power plant offers its limited energy (in comparison to the other generation plants) at 0 €/MWh, these 0 prices make its power very attractive and its included first in every auction. It does not have maximum or minimum power constraints, nor is it limited by any ramp-up or ramp-down limits.

Table 2, Generator 6, Windpower plant characteristics

$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
5MW	10MW	15MW	20MW	10MW

The three demand points D1, D2, D3 (indexed by i) will be modeled using the parameters from Table 3. Similar to the generators in Table 1, these also offer multiple demand blocks, in this case three (indexed by k), each with its power in MW and associated price in €/MWh.

Table 3, Consumers Characteristics

Demand units	Bus	$P_{D_{ikt}}^b [MW]$			$\lambda_{D_{ikt}}^b [€/MWh]$		
		$k=1$	$k=2$	$k=3$	$k=1$	$k=2$	$k=3$
$t=1$	$i=1$	1	0	0	100	0	0
	$i=2$	2	100	100	40	100	80
	$i=3$	3	80	50	30	100	65
$t=2$	$i=1$	1	1	0	60	0	0
	$i=2$	2	80	80	32	60	48
	$i=3$	3	64	40	24	60	39
$t=3$	$i=1$	1	2	0	120	0	0
	$i=2$	2	170	170	68	120	96
	$i=3$	3	136	85	51	120	78
$t=4$	$i=1$	1	2	0	140	0	0
	$i=2$	2	130	130	52	140	112
	$i=3$	3	104	65	39	140	91
$t=5$	$i=1$	1	2	0	130	0	0
	$i=2$	2	140	140	56	130	104
	$i=3$	3	112	70	42	130	85

Finally, the five transmission lines are modeled using the data from Table 4. These data complete this DC-OPF model with reactance x_{kl} [p.u.], line capacity S_{max} [MW], and base power $sb=100$ [MW].

These lines will allow each bus to share excess power or excess demand with other buses

Table 4, Transmission lines Characteristics

Transmission lines ($sb = 100MW$)			
$k =$	$l =$	x_{kl} [p. u]	S_{kl}^{max} [MW]
1	2	0.058	150
1	3	0.04	150
2	3	0.08	150
2	4	0.08	150
3	4	0.12	150

B. Presenting mathematical equations (TCMPA transmission constrained multiperiod auction based equations)

The optimization objective is to maximize Social Welfare (SW) within every node (bus) where all powers (generated or demanded) are positive variables.

Equation 1, Social welfare maximization.

$$\max_{P_G, P_D, \lambda} SW(P_G, P_D) = \sum_{t \in \mathcal{T}} \left[\sum_{i \in \mathcal{D}} \sum_{k \in \mathcal{B}_D^b} \lambda_{D_{ikt}}^b P_{D_{ikt}} - \sum_{i \in \mathcal{G}} \sum_{k \in \mathcal{B}_G^b} \lambda_{G_{ikt}}^b P_{G_{ikt}} \right]$$

That SW is subject to the following rules Equation2

Equation 2, Matched generation and demand

$$\begin{aligned} P_{G_{ikt}} &\leq P_{G_{ikt}}^b & k \in \mathcal{B}_G^b, i \in \mathcal{G}, t \in \mathcal{T} \\ P_{D_{ikt}} &\leq P_{D_{ikt}}^b & k \in \mathcal{B}_D^b, i \in \mathcal{D}, t \in \mathcal{T} \\ P_{G_{it}}^T &= \sum_{k \in \mathcal{B}_G^b} P_{G_{ikt}} & i \in \mathcal{G}, t \in \mathcal{T} \\ P_{D_{it}}^T &= \sum_{k \in \mathcal{B}_D^b} P_{D_{ikt}} & i \in \mathcal{D}, t \in \mathcal{T} \end{aligned}$$

The generation ramp up/down limits are modeled with Equation3.

Equation 3

$$\begin{aligned} P_{G_{it}}^T - P_{G_{i(t-1)}}^T &\geq -R_{G_i}^{down} & i \in \mathcal{G}, t \in \mathcal{T} \\ P_{G_{it}}^T - P_{G_{i(t-1)}}^T &\leq R_{G_i}^{up} & i \in \mathcal{G}, t \in \mathcal{T} \end{aligned}$$

The generation maximum/minimum limits translate to equation4.

Equation 4

$$\begin{aligned} P_{G_{it}}^T &\geq P_{G_i}^{min} u_{it} & i \in \mathcal{G}, t \in \mathcal{T} \\ P_{G_{it}}^T &\leq P_{G_i}^{max} u_{it} & i \in \mathcal{G}, t \in \mathcal{T} \end{aligned}$$

As there is no energy storage modeled in our grid, the power generated must match the power consumed in every node Equation5 (+imports/exports).

Equation 5, Internodal market equilibrium

$$\begin{aligned} \sum_{(k,l) \in \mathcal{L}} P_{klt} - \sum_{(l,k) \in \mathcal{L}} P_{lkt} \\ = \sum_{i \in \mathcal{G}_k} P_{G_{it}}^T - \sum_{i \in \mathcal{D}_k} P_{D_{it}}^T \\ k \in \mathcal{B}, t \in \mathcal{T} \end{aligned}$$

In order to model the power flow through transmission lines, active power flow relates to grid frequency differences between buses Equation6 (frequency reference in bus1 = 0).

Equation 6, line power flows

$$P_{klt} = sb \cdot (\theta_{kt} - \theta_{lt}) / x_{kl}$$

$$(k, l) \in \mathcal{L}, t \in \mathcal{T}$$

Transmission line saturation is modeled with Equation 7.

Equation 7, transmission line maximum power flow.

$$-S_{kl}^{max} \leq P_{klt} \leq S_{kl}^{max}$$

$$(k, l) \in \mathcal{L}, t \in \mathcal{T}$$

III. NUMERICAL RESULTS

GAMS was configured to find an optimal solution with 0.0

gap, this guarantees $X^*_{\text{gams}} \equiv X^*_{\text{tcampa.0}}$

The interesting parameters from the GAMS found equilibrium are going to be:

- Market clearing (power and price) for each t
- Merchandizing surplus each t (power flows)
- Nodal clearings each node each t
- All different social welfare results (TCSW* SW* and NSW*) each t

A. Results for the base case test T=1

There are no saturated Transmission lines so the whole market was able to clear at the same price of 35€/MW

Table 5, t=1 every transmission line power and price

INDEX 1 = t1	energy	price sell	price buy
bus1.bus2	113.923	35.000	35.000
bus1.bus3	135.077	35.000	35.000
bus2.bus3	15.055	35.000	35.000
bus2.bus4	66.022	35.000	35.000
bus3.bus4	33.978	35.000	35.000

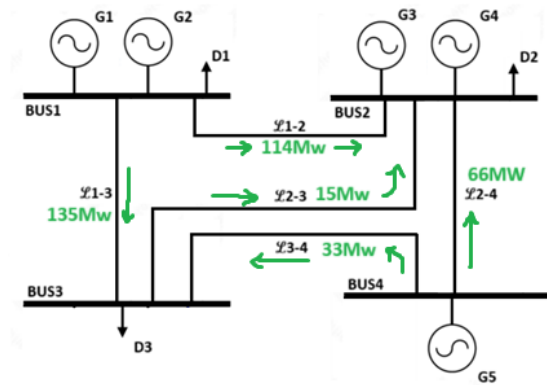


Figure 2, Power flow between nodes at t=1

TCSW* was maximized to 22890.

Table 6, TCSW* for all T, t=1 highlighted in yellow

---	361	PARAMETER	market_equilibrium_T							
	energy	gen.	dem.	inflow	outflow	surp. gen.	surp. dem.	SW		
t1	759.055	395.000	395.000	364.055	364.055	3525.000	19365.000	22890.000		
t2	865.630	305.000	305.000	260.630	260.630	-1295.000	9503.000	8008.000		
t3	922.750	480.000	480.000	432.750	432.750	23250.000	11532.800	35222.800		
t4	872.138	483.000	483.000	389.138	389.138	7501.848	24866.728	32370.576		
t5	911.642	484.836	484.836	426.805	426.805	6899.772	24099.216	30988.988		

In **bus1** some limitations were hit, for example G2 bottomed its Pmin of 28Mw so it was forced to sell unmatched power at a loss of 5€/MW.

G1 also hit its Pramp up limit of 100MW/t limiting its matched generation to 250Mw.

Due to D1 offering to buy just 1MW almost all energy matched in bus1 was sold to other buses.

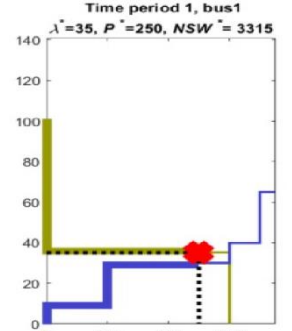


Figure 3, market clearing for bus1 t=1

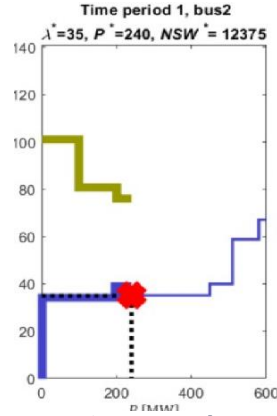


Figure 4, market clearing for bus2 t=1

With **bus2** having a very hungry D2, it would make sense for G3 and G4 to have all their generation matched, unluckily for those operators their bid was too expensive so energy imports from all other cheaper generators were in order.

Bus2 becomes an absolute energy importer in t=1 (Surplus generation -225)

Both G3 and G4's Rampdown limits were reached having to sell their energy also at 5€/MW loss.

Bus3 has no generation, and its demanded prices are very attractive for all other nodes, that's why all its demand is matched with a virtual generation block.

The only exception are 15Mw that are using Bus3 as a buffer to channel generation from buses1 and 4 and routing it to bus2.

After all the optimization bus3 is setting the clearing price of the whole grid at 35€/Mw

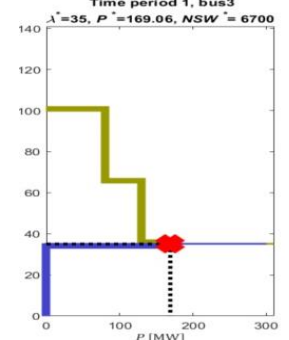


Figure 5, market clearing for bus3 at t=1

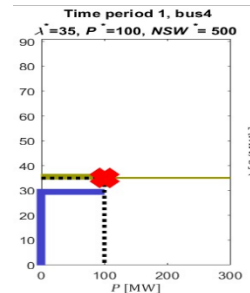


Figure 6, market clearing for bus4 at t=1

Bus4 has only one generator so it makes sense all its energy bidding was matched and exported to other nodes.

Luckily for G5 operator the internodal market cleared the price higher than its offered price so its earning +5€ more per Mw.

B. Results for the base case test T=2.

In a similar manner than T=1, transmission lines are not yet saturated so all nodes cleared to the same market price of 21€/Mw

Table 7, t=2 every transmission line power and price

INDEX 1 = t2	energy	price sell	price buy
bus1.bus2	90.718	21.000	21.000
bus1.bus3	108.282	21.000	21.000
bus2.bus3	11.630	21.000	21.000
bus2.bus4	34.652	21.000	21.000
bus3.bus4	15.348	21.000	21.000

The power flow remains the same as t=1 with slightly lower values.

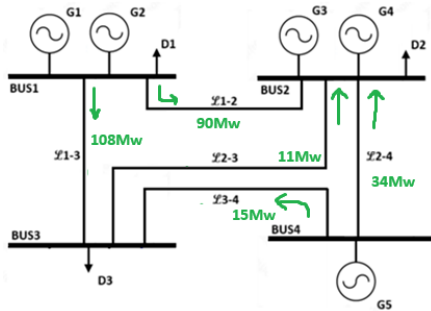


Figure 8, Power flow between nodes in t=2

The overall demand plumbed and all the generation units were selling at a loss (surplus generation -1295) because they were all hitting their ramp-down and Pmin limitations, that's why the low TCSW* =80008.

Table 8, TCSW* for all T, t=2 highlighted in yellow

361 PARAMETER market_equilibrium_T									
	energy	gen.	dem.	inflow	outflow	surp. gen.	surp. dem.	SW	
t1	759.055	395.000	395.000	344.055	344.055	3525.000	19365.000	22890.000	
t2	365.630	305.000	305.000	240.630	240.630	-1295.000	8903.000	8008.000	
t3	922.750	490.000	490.000	452.750	452.750	23290.000	11932.000	35222.000	
t4	875.138	485.000	485.000	359.138	359.138	7501.848	24868.728	32370.576	
t5	911.642	484.836	484.836	426.805	426.805	6889.772	24099.216	30989.988	

Back to **bus1**, D1 is still demanding just a mere 1Mw, that makes G1 and G2 to be next exporters again.

G2 is still rock bottom at its Pmin 28Mw and now loosing even more money than before -9€/Mw.

G1 with its cheap energy offers capped its maximum Power of 300Mw and got a matching bid for all.

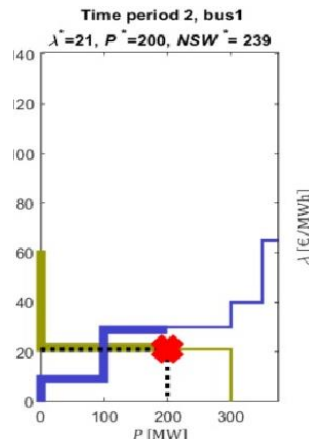
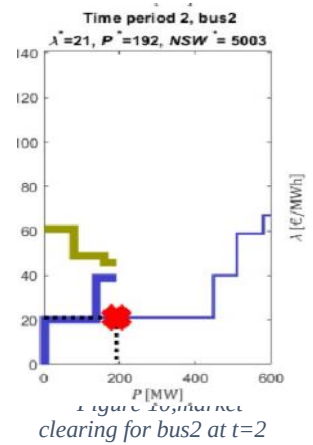


Figure 9, market clearing for bus1 at t=2

Things in **bus2** are looking very similar, G3 and G4 bids are still very expensive, they still managed for some of their cheapest offers to be accepted. That makes bus2 to be still a net energy importer from every other cheaper energy sources.



Bus3, our demand hub is still marking the whole grid clearing price at 21€/Mw.

It is still used as a buffer for a small 11Mw going to bus2 ([1] Three agent transaction.)

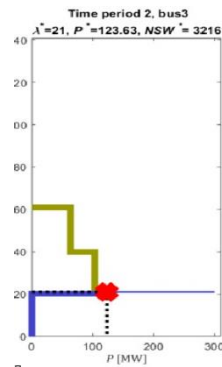


Figure 7, market clearing for bus3 at t=2

Bus4 with its lonely G5, is still exporting all of its energy but unluckily the market cleared with a cheap energy price, G5's operator is losing -9€/Mw in this situation.

G5 is hitting its Rampdown generation of -50Mw/t.

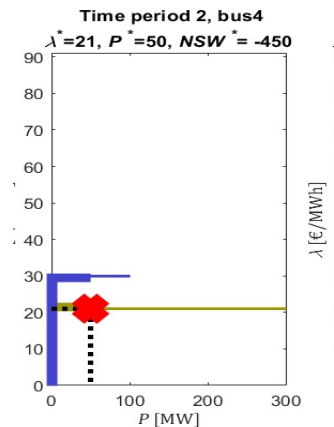


Figure 11, market clearing for bus4 at t=2

C. Results for the base case test $T=3$.

Transmission lines are starting to be saturated, and the demand is suddenly increasing; this is taking away some of the power the demand unit D3 had over the whole market.

[1]Links which operate below their line-capacity may still have nodal price differences (congestion is not a line, it's a network)

Table 9, $t=3$ every transmission line power and price, saturated line in red.

INDEX 1 = t3	energy	price sell	price buy	mer. surp.
bus1.bus2	150.000	90.000	69.600	3060.000
bus1.bus3	148.000	78.000	69.600	1243.200
bus2.bus3	34.750	90.000	78.000	417.000
bus2.bus4	73.900	90.000	85.200	354.720
bus3.bus4	26.100	78.000	85.200	-187.920

The power flow almost doubles since $t=2$, and now the line from bus1 to bus2 is saturated, with the line from bus1 to bus3 almost saturated aswell.

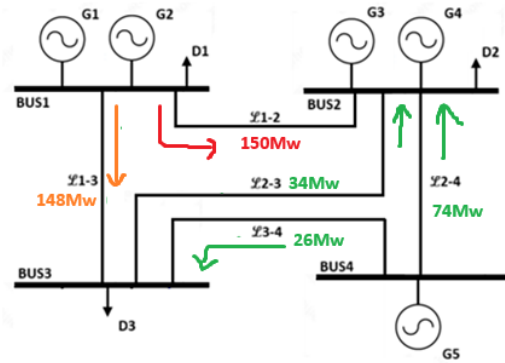


Figure 12, Power flow between nodes in $t=3$, saturated line in red

The market energy sharing between nodes is limited so energy price clearings are not the same between nodes anymore. It is starting to look more like a real market, TCSW* is higher because generation and demand sides are starting to reach compromises. (Previously the demand side was getting away with cheap energy prices and generators selling at loss).

Table 10, TCSW* for all T , $t=3$ highlighted in yellow

961 PARAMETER market_equilibrium_T							
	energy	gen.	dem.	inflow	outflow	surp. gen.	surp. dem.
t1	759.055	395.000	395.000	364.055	364.055	3525.000	19365.000
t2	565.630	305.000	305.000	260.630	260.630	-1295.000	9309.000
t3	922.750	490.000	490.000	432.750	432.750	23290.000	11932.800
t4	872.138	489.000	489.000	389.138	389.138	7501.848	24846.720
t5	911.642	484.836	484.836	426.805	426.805	6889.772	24099.216

Back again to **bus1**, D1 still demanding negligible power, G1 and G2 are facing a sudden rise in demand from other buses.

The bad news for G1 is they reached their Prampup limit of 100Mw/t and they are already saturating all transmission lines, there is no space for any of the more expensive G2 power offers.

The good news for G1 operator is that they are earning a whopping +39€/Mw more than they were bidding for.

Even G2 with no bids matched yet is getting about +29€/Mw for their Pmin limited generation.

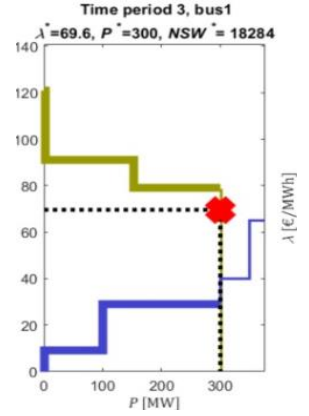


Figure 13, market clearing for bus1 at $t=3$

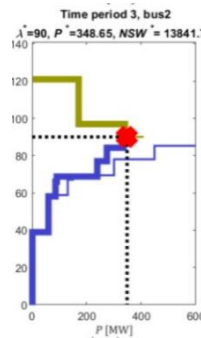


Figure 14, market clearing for bus2 at $t=3$

Bus3, because of the congested line proving cheap energy from bus1, they are paying top euro price for some of the most expensive generation bids from alternative energy sources.

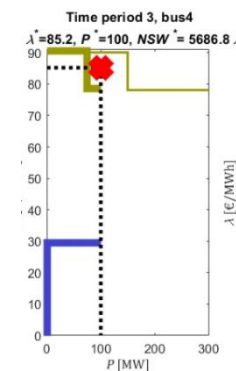


Figure 16, market clearing for bus4 at $t=3$

In **bus2**, G3 and G4's expensive energy bids are starting to be matched.

They are both hitting their Prampup limits and earning some €/Mw

Bus2 is the one setting the most expensive clearing price of all, as there is a lot of demand now their expensive prices are being met.

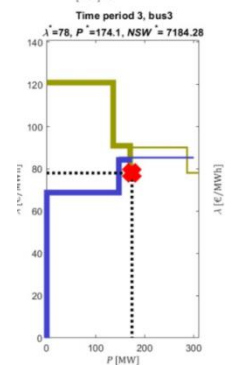


Figure 15, market clearing for bus3 at $t=3$

Generation operators in **Bus4**, they are the ones earning a lot of money now, G5 hits its Prampup of +50Mw and manages to sell all the energy that is able to produce at +55€/Mw to other nodes.

D. Results for the base case test $T=4$.

Transmission lines are still saturated, but there is a slight shift in line flow, the saturated line is now the one from bus1 to bus3, the overall market demand is still high but not that much.

It is also worth noticing, merchandizing surplus in both buses going to bus2 is negative, this signals inefficiencies in the market, due to lower demand and Transmission line constraints.

Table 11, $t=4$ every transmission line power and price, saturated line in red.

INDEX 1 = t4				
	energy	price sell	price buy	mer. surp.
bus1.bus2	124.138	60.295	30.000	3760.794
bus1.bus3	150.000	90.143	30.000	9021.429
bus2.bus3	15.000	60.295	90.143	-447.714
bus2.bus4	66.000	60.295	72.234	-787.976
bus3.bus4	34.000	90.143	72.234	608.890

The power flow panorama is very similar to the one in $t=3$, power flow directions hadn't changed since the start of this simulation.

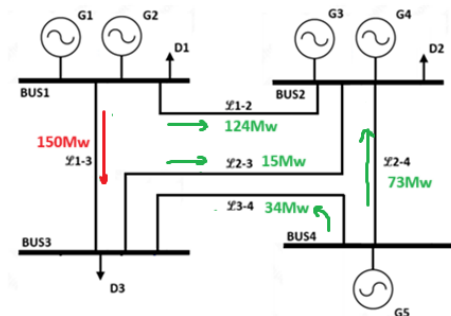


Figure 17, Power flow between nodes in $t=4$, saturated line in red

TCSW* is still high but starting to loosen due to drop in demand. Is also worth notice how the generation operators' benefits are dropping (Surplus generation = 7500, compared to 3x times in $t=3$)

Table 12, TCSW* for all T , $t=4$ highlighted in yellow

361 PARAMETER market_equilibrium_T									
	energy	gen.	dem.	inflow	outflow	surp. gen.	surp. dem.	SW	
t1	759.055	395.000	395.000	364.055	364.055	3925.000	19365.000	22890.000	
t2	565.630	305.000	305.000	260.630	260.630	-1295.000	9303.000	8008.000	
t3	922.750	490.000	490.000	432.750	432.750	23390.000	-11932.800	35222.800	
t4	872.138	493.000	493.000	389.138	389.138	1501.845	24848.728	32370.878	
t5	911.642	494.836	494.836	426.805	426.805	6889.772	24099.216	30989.988	

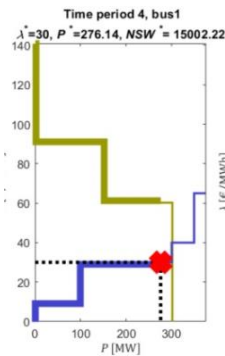


Figure 18, market clearing for bus1 at $t=4$

In bus2, G3 and G4 bids are still being met but the demand didn't change much so they are not hitting any of their Pramp limitations at the moment.

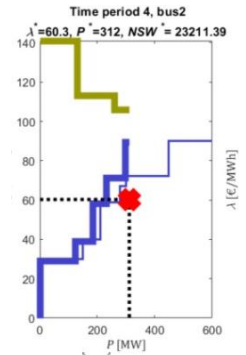


Figure 19, market clearing for bus2 at $t=4$

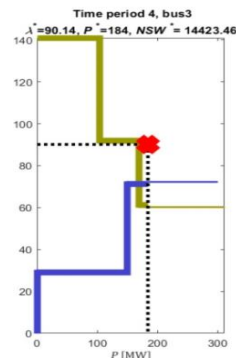


Figure 20, market clearing for bus3 at $t=4$

Bus4, also remains similar, the optimization TCMPA is still dealing a good price for G4, now earning a 42€/Mw more than the price they were bidding.

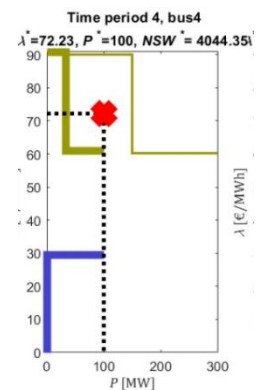


Figure 21, market clearing for bus4 at $t=4$

In Bus1, G1 and G2 scenario remains the same, G1 is matching most of its power bids and there is no transmission line resource left for G2 to include its power.

This time in bus1, G1 only allocated 270Mw of its 300Mw available.

Bus3 D3 remains as the one paying the most expensive Mw.

The transmission line joining bus3 with its cheaper generation source bus1 is saturated so nodal market clearing is influenced by bus4 expensive imports.

E. Results for the base case test $T=5$.

Not much to comment, same line saturated than in $t=4$.

Same generation and demand volume than in $t=4$, and slight increase in Transmission line usage.

Same market inefficiencies signaled by negative merch surplus in bus2.

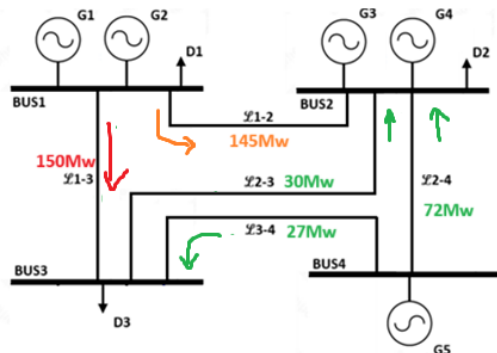
Table 13, $t=5$ every transmission line power and price, saturated line in red.

INDEX 1 = t5

	energy	price sell	price buy	mer. surp.
bus1.bus2	145.974	57.705	30.000	4044.172
bus1.bus3	150.000	85.000	30.000	8250.000
bus2.bus3	30.831	57.705	85.000	-841.548
bus2.bus4	72.333	57.705	68.623	-789.735
bus3.bus4	27.667	85.000	68.623	453.115

Bus1 is still saturating its Transmission lines trying to export all its energy.

Table 14, Power flow between nodes in $t=5$, saturated line in red



TCSW* is very similar to previous cases, dropping a bit due to slightly higher usage of Transmission lines.

Table 15, TCSW* for all T , $t=5$ highlighted in yellow

361 PARAMETER market_equilibrium_T								
	energy	gen.	dem.	inflow	outflow	surp. gen.	surp. dem.	S
t1	759.055	395.000	395.000	364.055	364.055	3525.000	19365.000	22890.00
t2	565.630	305.000	305.000	260.630	260.630	-1295.000	8303.000	8008.00
t3	922.750	490.000	490.000	432.750	432.750	23290.000	11932.800	35222.80
t4	872.138	483.000	483.000	389.138	389.138	7501.848	24868.728	32370.57
t5	911.642	484.836	484.836	426.805	426.805	6889.772	24099.216	30988.98

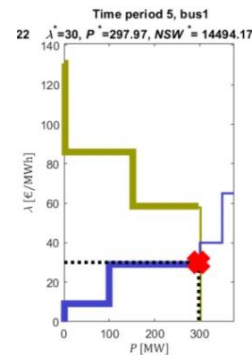


Figure 22, market clearing for bus1 at $t=5$

Bus2 clears its market fulfilling all demand from D2, matching the cheapest biddings from local generators G3 G4, importing everything else from bus4 and bus3 (proxy of bus1)

Practically all generation bids from G1 were matched in bus1 this time, this means G1 is almost hitting its Pmax again and both L1-2 and L1-3 are saturated again.

G2 still remains out of the bidding providing its usual Pmin=28Mw.

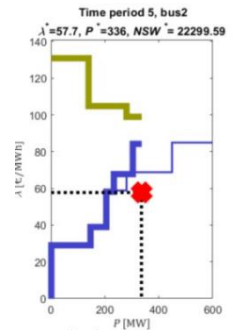


Figure 23, market clearing for bus2 at $t=5$

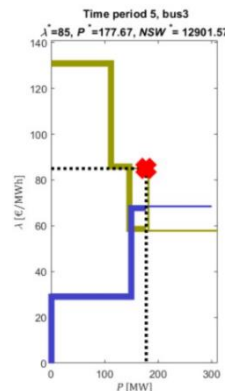


Figure 24, market clearing for bus3 at $t=5$

Bus3, as usual paying the most from all energy imports, as the market's transmission lines are oversaturated. Clearing the market far away from the optimal point, this indicates heavy inefficiencies in the market energy allocation.

In bus4 G5 as usual matching its only bid, this time the market inefficiency is leaning heavily against the consumers.

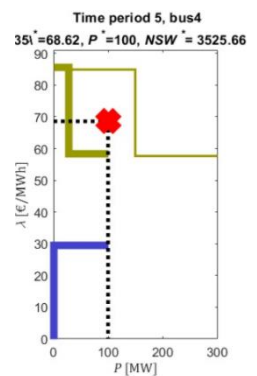
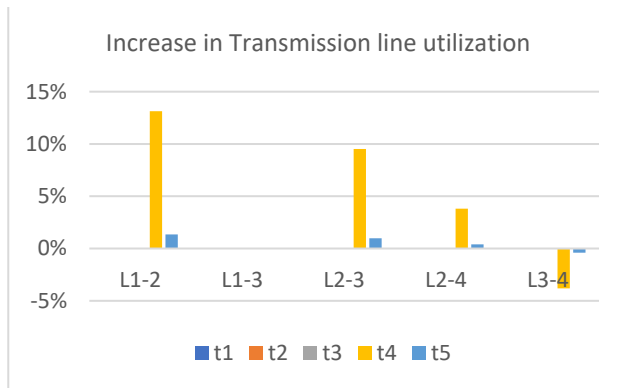


Figure 25, market clearing for bus4 at $t=5$

F. Results and Analysis for the case with the inclusion of the Wind turbine

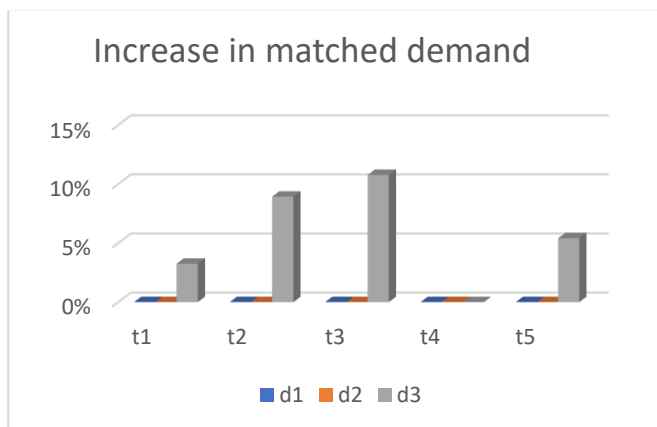
Launching the same optimization problem but now adding a low power renewable generator G6 located in D3.

By comparing against the base case (no WPP), the following graphs will help us identify in who's interest is the addition of this new Wind power plant.



There is an overall increase in the utilization of all not saturated lines.

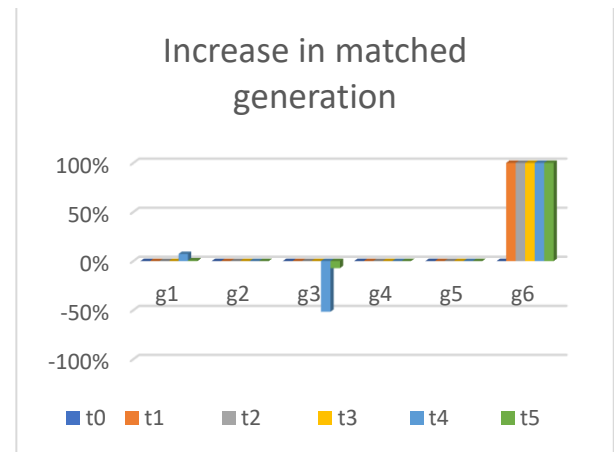
The addition of the WPP didn't help clear any of the bottlenecks (congestion taxes), in this case the TSO would not be interested in the WPP unless higher power turbines are installed.



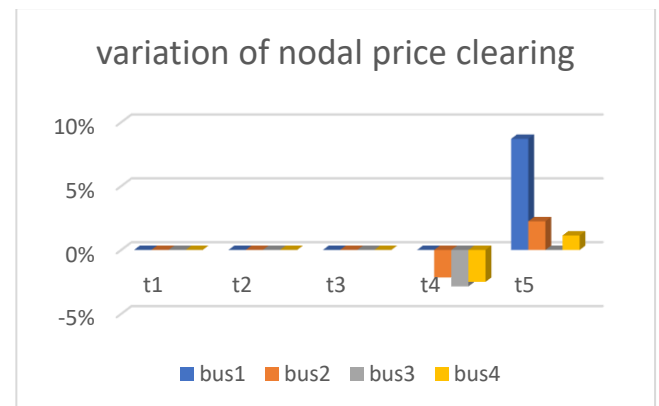
The addition of WPP served only to cover D3 huge power demand, this means all clean energy produced by the WPP was spent locally in bus3.

It would be in the Demand side's interest to install this WPP, specially D3, and indirectly the other demand actors.

Leaning on the side of the generators, G1 would be a bit interested in the construction of the new WPP, but G3 on the other hand will lose a big chunk of its matched power if the new WPP is put in place.

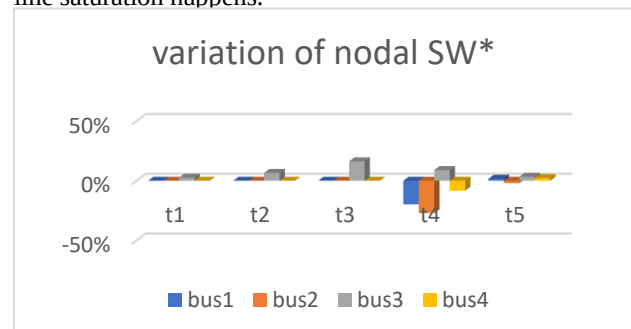


The impact of the WPP is very low when the transmission lines are not congested due to having an order of magnitude lower power offer than the other generator counterparts, but it is big enough to clear a couple of bottlenecks in t4 and t5, specially affecting bus1 clearing price.



What about nodal SW* our optimization parameter.

Bus3 is greatly benefited since start by having the local WPP, it is not the case of every other bus when Transmission line saturation happens.



IV. CONCLUSIONS

As an initial conclusion, it is always advantageous for a community of consumers without local energy generation to be able to install a wind power plant (WPP) to supply at least part of their local consumption.

Studying the global impact of such a WPP on the network of 4 buses, it makes sense to construct it to alleviate an already congested grid. However, it would be necessary to build a WPP with greater capacity to effectively relieve the congestion on the L1-3 line.

The results indicate that the installation of the WPP is in the interest of D3. However, there are many generating actors who benefit from the line congestion, and thus it is not in their economic interest to support the construction of this new plant.

If these generating actors had a say in the decision, it would make sense for them to oppose the construction of the WPP.

As a detail, a simulation of a WPP with 30 times the installed capacity was tested, and even with this power, the line congestion was not alleviated. Therefore, it would be more effective to opt for the installation of more distributed generation plants, particularly at bus 3 and bus 2, which are the main energy consumers.

V. REFERENCE

- [1] "Nodal prices and transmission rights: A critical appraisal" by Shmuel S. Oren, Pablo T. Spiller, Pravin Varaiya and Felix Wu (The Electricity Journal, Volume 8, Issue 3, April 1995, Pages 24–3)